Safety Data Sheet
ABS

Compiled according to GB/T 16483, GB/T 17519
Product Reference code:UM00001
Issue date: 2023/11/10 Version: 1.0

SECTION 1 Chemical product and company identification

Chemical Chinese name	:	ABS (绿色, 黑色, 白色, 橙色, 蓝色, 红色, 黄色, 银, 灰色, 珍珠 金)
Chemical English name	:	ABS (Green, Black, White, Orange, Blue, Red, Yellow, Silver, Grey, Pearl Gold)
Name of company	:	UltiMaker
Title	:	Supplier
Address	:	The Netherlands Geldermalsen Watermolenweg 2
Zip code	:	4191 PN
Tel.	:	+31 (0) 88 383 4000 (9 AM - 5 PM CET)
E-mail	:	Product-Compliance@Ultimaker.com
Emergency number	:	+31 (0) 88 383 4000 (during office hours: 9 AM - 5 PM CET)
Recommended use of the chemical	:	3D-Printer filament
Restricted use of the chemical	:	This product must not be used in applications other than those identified above, without first seeking advice of the supplier

SECTION 2 Hazards identification

Emergency overview

Appearance: Filament, Various colours. Keep away from: Strong oxidizing agents. Non flammable. Not rapidly degradable. The product is non-reactive under normal conditions of use, storage and transport

To our knowledge, this product does not present any particular risk, provided it is handled in accordance with good occupational hygiene and safety practice

GHS hazard classification

Other hazards not mentioned above are Not applicable or No data is available.

Label elements

No data available

Physical and chemical hazards

No additional information available

Health hazards

Symptoms/effects	:	No acute and delayed symptoms and effects are observed
Symptoms/effects after skin contact	:	Risk of thermal burns on contact with molten product

Environmental hazards

No additional information available

Other hazards

Risk of thermal burns on contact with molten product

SECTION 3 Composition/information on ingredients

Product form : Mixture.

Ingredient(s)	Concentration or concentration ranges (w/w %)	CAS No.
2-Propenenitrile, polymer with 1,3-butadiene and ethenylbenzene		9003-56-9
Poly(ethylene terephthalate)		25038-59-9
Polycarbonate		25037-45-0
Titanium dioxide (Additive for ABS White, Blue, Green, Grey, Yellow, Orange)	< 1.5	13463-67-7
Carbon black (Additive for ABS Black, Grey, Orange)	< 1.5	1333-86-4

SECTION 4 First-aid measures

Description of necessary first-aid measures

First-aid measures general : If you feel unwell, seek medical advice (show the label where possible)

First-aid measures after inhalation : Remove person to fresh air and keep comfortable for breathing.

In molten state: Hazardous vapours may be released

First-aid measures after skin contact : Wash skin with plenty of water and soap.

Take off contaminated clothing.

In case of contact with molten product, cool rapidly with water and seek immediate medical attention. Do not attempt to remove molten product from skin because skin will tear easily.

Burns caused by molten material must be treated clinically

First-aid measures after eye contact : Rinse eyes with water as a precaution.

In the event of contact with molten product: Immediately flush eyes thoroughly with water for at least 15 minutes.

Get immediate medical advice/attention.

First-aid measures after ingestion : Call a poison center or a doctor if you feel unwell

Most important symptoms/effects

Symptoms/effects : No acute and delayed symptoms and effects are observed

Symptoms/effects after skin contact : Risk of thermal burns on contact with molten product

Advices for first aid responders

No additional information available

Notes for the doctor

Other medical advice or treatment : Treat symptomatically

SECTION 5 Fire-fighting measures

Extinguishing media

Suitable extinguishing media	:	Use extinguishing media appropriate for surrounding fire: Water spray, Dry powder, Foam, Carbon dioxide
Unsuitable extinguishing media	:	Do not use a solid water stream as it may scatter and spread fire

Specific hazards

Hazardous decomposition products in case of fire	:	Under fire conditions, hazardous fumes will be present: Carbon dioxide, Carbon monoxide
Explosion hazard	:	Material can accumulate some static charge during transfer Prevent build-up of electrostatic charges (e.g, by grounding)

Advice for firefighters and protective measures

Firefighting instructions	:	No additional information available
Protection during firefighting	:	Do not attempt to take action without suitable protective equipment Self-contained breathing apparatus Complete protective clothing
Precautionary measures fire	:	Do not allow run-off from fire-fighting to enter drains or water courses

SECTION 6 Accidental release measures

Personal precautions, protective equipment and emergency procedures

Heat and ignition sources	:	Keep away from heat, sparks and flames Keep out of direct sunlight
General measures	:	No additional information available
Personal Precautions, Protective Equipment and Emergency Procedures	:	No additional information available

For non-emergency personnel

Protective equipment	:	Wear recommended personal protective equipment Refer to section 8.2 Remove contaminated clothing and shoes
Emergency procedures	:	None in particular In molten state: Do not breathe vapours Ventilate spillage area Avoid contact with skin, eyes and clothing

For emergency responders

Protective equipment	:	Do not attempt to take action without suitable protective equipment For further information refer to section 8: "Exposure controls/personal protection"

Environmental precautions

Avoid release to the environment

Notify authorities if product enters sewers or public waters

Methods and material for containment and cleaning up

Methods for cleaning	:	No additional information available
Methods and Equipment for Containment and Cleaning up	:	Sweep up and put in a closed container for disposal If melted: allow liquid to solidify before taking it up
For containment	:	No additional information available

Prevention measures for secondary accidents

Prevention Measures for Secondary Accidents	:	No additional information available
Other information	:	Dispose of materials or solid residues at an authorized site

SECTION 7 Handling and storage

Handling

Precautions for safe handling	:	Ensure good ventilation of the work station In molten state: Do not breathe vapours Avoid contact with skin, eyes and clothing Wear personal protective equipment
Hygiene measures	:	Handle in accordance with good industrial hygiene and safety practice Do not eat, drink or smoke when using this product. Always wash hands after handling the product Contaminated work clothing should not be allowed out of the workplace. Wash contaminated clothing before reuse.
Local and general ventilation	:	No additional information available

Storage

Storage conditions	:	To guarantee the quality and properties of the product: Store in a well-ventilated place. Store in original container Keep container tightly closed to avoid moisture absorption and contamination
Material used in packaging/containers	:	No additional information available
Incompatible materials	:	Strong oxidizing agents.
Storage temperature	:	15 – 25 °C (Relative air humidity: <50%)
Heat and ignition sources	:	Keep away from heat, sparks and flames Keep out of direct sunlight

SECTION 8 Exposure controls / Personal protection equipment

Occupational exposure limits

Titanium dioxide (Additive for ABS White, Blue, Green, Grey, Yellow, Orange) (13463-67-7)	
China - Occupational Exposure Limits	
Local name	二氧化钛粉尘 # Titanium dioxide dust

Titanium dioxide (Additive for ABS White, Blue, Green, Grey, Yellow, Orange) (13463-67-7)	
OEL TWA	8 mg/m³ 总尘
OEL chemical category (CN)	Possibly carcinogenic to humans dust
Catalogue of Occupational Hazard Factors	Category 1 - Dusts
Remark (CN)	G2B（对人可疑致癌 (Possibly carcinogenic to humans)）
Regulatory reference	GBZ 2.1-2019

Carbon black (Additive for ABS Black, Grey, Orange) (1333-86-4)	
China - Occupational Exposure Limits	
Local name	炭黑粉尘 # Carbon black dust
OEL TWA	4 mg/m³ 总尘
Remark (CN)	G2B（对人可疑致癌 (Possibly carcinogenic to humans)）
Regulatory reference	GBZ 2.1-2019

Exposure limit values for the other components

Biological limit values

No additional information available

Monitoring methods

No additional information available

Appropriate engineering controls

Use process enclosures, local exhaust ventilation or other engineering controls to keep airborne levels below specified exposure limits. If user operations generate dust, fumes or mist, use ventilation to keep exposure to airborne particles below the exposure limit.

Ventilation conditions (1 printer): Provide a good standard of general ventilation, not less than 2 air changes per hour (assumes a room volume of: 30 m³)

Personal protective equipment

Thermal hazard protection : Risk of thermal burns on contact with molten product
Hazardous vapours may be released
In molten state: Wear respiratory protection/heat resistant gloves

Environmental exposure controls : Avoid release to the environment.

Other information : Handle in accordance with good industrial hygiene and safety procedures
Do not eat, drink or smoke when using this product.
Wash hands immediately after handling the product
Take off contaminated clothing and wash before reuse.

Hand protection : None under normal conditions
Use insulated gloves when handling this material hot

Type	Material	Permeation	Thickness (mm)	Penetration	Standard
In molten state:, Chemically resistant protective gloves, Heat-resistant.	Nitrile rubber (NBR), Natural rubber, Butyl rubber, Polyvinylchloride (PVC).	6 (> 480 minutes).	>0.35		EN 374, EN 407.

Eye protection : None under normal use
In molten state: Wear eye protection

Type	Use	Characteristics	Standard
Safety glasses with side shields.	In molten state.		EN 166.

Skin and body protection : None under normal use
In molten state: Wear suitable protective clothing

Type	Standard
Long sleeved protective clothing.	EN 13688.

Respiratory protection : None under normal use
In molten state: In case of insufficient ventilation, wear suitable respiratory equipment

Device	Filter type	Condition	Standard
In molten state:, Half-face mask.	Type B/P2.		EN 140, EN 14387.

SECTION 9 Physical and chemical properties

Physical state : Solid
Appearance : Filament
Colour : Various colours
Odour : Neutral
pH : No data available
Melting point : 140 – 170 °C
Freezing point : Not applicable
Boiling point : Not applicable
Flash point : Not applicable
Auto-ignition temperature : > 400 °C
Decomposition temperature : > 280 °C.
Flammability : Non flammable
Vapour pressure : No data available
Relative vapour density at 20°C : No data available
Density : 1.1 g/cm³

Solubility	:	No data available
Solubility in water	:	Insoluble
Partition coefficient n-octanol/water (Log Pow)	:	No data available
Viscosity, kinematic	:	Not applicable
Explosive limits (vol %)	:	Not applicable
Lower explosive limit (LEL)	:	No data available
Upper explosive limit (UEL)	:	No data available
Radioactive	:	No
Particle size distribution	:	Not applicable

SECTION 10 Stability and reactivity

Chemical stability	:	Stable under normal conditions
Reactivity	:	The product is non-reactive under normal conditions of use, storage and transport
Possibility of hazardous reactions	:	No dangerous reactions known under normal conditions of use
Conditions to avoid	:	None under recommended storage and handling conditions (see section 7). Printing process: Avoid temperature above 260°C
Incompatible materials	:	Strong oxidizing agents
Hazardous decomposition products	:	Under normal conditions of storage and use, hazardous decomposition products should not be produced Under fire conditions, hazardous fumes will be present: Carbon dioxide, Carbon monoxide
Other properties	:	No additional information available

SECTION 11 Toxicological information

Acute toxicity

Acute toxicity (oral)	:	Not classified
Acute toxicity (dermal)	:	Not classified
Acute toxicity (inhalation)	:	Not classified

Skin corrosion/irritation

Skin corrosion/irritation	:	Not classified

Serious eye damage/eye irritation

Serious eye damage/irritation	:	Not classified

Respiratory or skin sensitisation

Respiratory or skin sensitisation	:	Not classified

Germ cell mutagenicity

Germ cell mutagenicity : Not classified

Carcinogenicity

Carcinogenicity : Not classified

Reproductive toxicity

Reproductive toxicity : Not classified

STOT - single exposure

STOT-single exposure : Not classified

STOT - repeated exposure

STOT-repeated exposure : Not classified

Aspiration hazard

Aspiration hazard : Not classified

ABS (Green, Black, White, Orange, Blue, Red, Yellow, Silver, Grey, Pearl Gold)	
Viscosity, kinematic	Not applicable
Density	1.1 g/cm³

SECTION 12 Ecological information

Ecotoxicity

Ecology - general : The product is not considered harmful to aquatic organisms nor to cause long-term adverse effects in the environment.

Hazardous to the aquatic environment, short–term (acute) : Not classified

Hazardous to the aquatic environment, long–term (chronic) : Not classified

Titanium dioxide (Additive for ABS White, Blue, Green, Grey, Yellow, Orange) (13463-67-7)	
LC50 fish 1	> 1000 mg/l

Persistence and degradability

ABS (Green, Black, White, Orange, Blue, Red, Yellow, Silver, Grey, Pearl Gold)	
Persistence and degradability	Not rapidly degradable

Bioaccumulative potential

No additional information available

Mobility in soil

No additional information available

Other adverse effects

Classification procedure (Ozone) : No data available

SECTION 13 Disposal considerations

Waste treatment methods : Dispose of contents/container in accordance with licensed collector's sorting instructions.

Contaminated container and packaging : No additional information available

Additional information : No additional information available

Product/Packaging disposal recommendations : Empty containers should be taken for recycling, recovery or waste in accordance with local regulation

Regional legislation (waste) : Dispose of in accordance with relevant local regulations

SECTION 14 Transport information

In accordance with JT/T 617 / IMDG / IATA

Overland transport (JT/T 617)	Transport by sea	Air transport
UN number		
Not regulated for transport		
Proper shipping name		
Not regulated	Not regulated	Not regulated
Transport hazard class(es)		
Not regulated	Not regulated	Not regulated
Packing group		
Not regulated	Not regulated	Not regulated
Environmental hazards		
Not regulated	Not regulated	Not regulated
No supplementary information available		

Special transport precautions

Overland transport (JT/T 617)

Not regulated

Transport by sea

Not regulated

Air transport

Not regulated

SECTION 15 Regulatory information

New Chemical Substance Environmental Management Registration Measures (MEE Order 12 of 2020)

Inventory of Existing Chemical Substances in China (IECSC) : Contains listed substance(s)

Acrylonitrile-butadiene-styrene terpolymer (CAS-No. 9003-56-9)
Poly(ethylene terephthalate) (CAS-No. 25038-59-9)
Polycarbonate (CAS-No. 25037-45-0)
Titanium oxide (CAS-No. 13463-67-7)
Carbon black (CAS-No. 1333-86-4)

Law of the People's Republic of China on the Prevention and Control of Occupational Diseases

Catalogue for Classification of Hazardous Factors of Occupational Diseases : Contains listed substance(s)

Titanium dioxide dust (CAS-No. 13463-67-7)
Carbon black dust (CAS-No. 1333-86-4)

SECTION 16 Other information

SDS Reason for revision : Not applicable

Abbreviations and acronyms

CAS-No.	Chemical Abstract Service number
CAS	Chemical Abstract Service number
EN	European Standard
GHS	Globally Harmonized System of Classification and Labelling of Chemicals
IATA	International Air Transport Association
IMDG	International Maritime Dangerous Goods
OEL	Occupational Exposure Limit
SDS	Safety Data Sheet
TWA	Time Weighted Average

Training advice : Ensure staff are informed of and trained on the nature of exposure and basic actions to minimise exposure.

Safety Data Sheet (SDS), China

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